



Lingyu Zhu

**Department of Computer Science,  
City University of Hong Kong**  
5006, 5/F, Run Run Shaw Creative  
Media Center, 18 Tat Hong Avenue,  
Kowloon Tong,  
Hong Kong  
**E-mail:** [lingyzhu-c@my.cityu.edu.hk](mailto:lingyzhu-c@my.cityu.edu.hk)  
**URL:** <https://github.com/lingyzhu0101>

## Research Statement Introduction

My current research is dedicated to advancing techniques for scene understanding and generation tasks in human-machine collaborative vision. One line of my research focuses on generative image and video compression for human-machine collaboration vision systems, characterizing the scene understanding of machine perception, developing benchmarks for open-world fine-grained tasks, and generative methods for designing ultra-low bitrate image or video compression. Another line of my research focuses on learning intelligent image and video enhancement (including conditions such as low light, haze, dust, and rain) as well as the reasoning task of multimodal models, and quality assessment. Herein, I will provide a brief summary of my doctoral research, along with the two thrusts, and end with how these are a step toward my ultimate goal as a researcher, which is to develop intelligent algorithms capable of comprehending and interacting with the visual environment. This statement aims to facilitate a diverse array of applications across fields such as robotics, embodied intelligence, human-machine interaction, and graphics.

### **Thrust 1: Generative Compression for in Human-Machine Collaboration System**

Image and video compression techniques aim to represent video signals compactly, facilitating efficient transmission and storage. However, existing deep learning methods that focus solely on distortion often optimize the rate-distortion trade-off, resulting in visually unpleasing outcomes at low bitrates. These approaches frequently produce “artifact-laden” results characterized by issues such as blurring and excessive smoothing, which hinder human perception. Moreover, current learned image codecs often struggle to achieve optimal performance for machine vision applications. This limitation primarily stems from their emphasis on pixel-level fidelity rather than extracting essential semantic information crucial for machine perception. To address these challenges, we introduce an end-to-end architecture designed for a human-machine collaboration compression framework. Our work [1, 2, 3, 4, 5] contributes to the development of generative compression techniques for vision systems in human-machine collaboration, enabling deeper exploration. In [2], we introduce a dynamic adaptation method that produces a single bitstream suitable for both humans and machines. Additionally, we explore the rate-distortion-realism trade-off in the generative image compression task utilizing pretrained stable diffusion [1]. Generative models have also been employed in applications related to the human body [3] and human faces [4] [5]. This paradigm achieves significant compression ratios while effectively minimizing accuracy loss in downstream tasks.

Unlike prior works, our model not only focuses on pixel-level fidelity but also integrates generative information or the semantic details, enabling a more robust understanding of the visual content. This dual approach allows for improved performance in both compression and downstream tasks, ensuring high-quality representations that facilitate machine vision applications.

**Thrust 2: Intelligent Enhancement and Quality Assessment of Visual Data** Currently, road safety remains an important issue, with over 1.35 million people dying in road accidents each year and 50 million injured. According to data provided by the Transportation Department in China, there were a total of 17,000 traffic accidents reported in 2023, which represents a 13% increase compared to 2022. Adverse weather and poor lighting conditions pose challenges for drivers in quickly identifying potential risks, such as pedestrians in close proximity, and accurately assessing safe distances and speeds. These factors significantly degrade driving performance. However, there is a lack of effective solutions specifically designed to monitor weather and lighting conditions at the road segment level for drivers. Existing engineering practices in road safety mainly focus on driver behavior and road design, and there is a significant gap between the growing need to explore fine-grained environmental conditions for enhanced driving safety and the availability of practical solutions. In response to the aspiration to improve road safety, we propose to deliver a fine-grained, real-time AI-powered solution. The developed system can address the challenges posed by extreme weather conditions. We aim to develop a visual enhancement model specifically focused on improving low-light enhancement capabilities. The proposed method outperforms both traditional and learning-based low-light enhancement algorithms. Compared to learning-based methods, the proposed methods achieve a 10% improvement in PSNR and SSIM. These criteria help ensure that the visual enhancements effectively enhance the perception of potential risks, contributing to safer driving conditions.

Compared to low-light image enhancement [6] [7] [8], enhancing low-light videos is more difficult due to the intertwined effects of noise, exposure, and contrast in the spatial domain, jointly with the need for temporal coherence. It is noted that we utilize the semantic priors of any region to design the diffusion model for low-light image enhancement in [8]. This approach allows us to leverage contextual information, improving the model’s ability to restore details and enhance visibility in challenging lighting conditions. By integrating semantic understanding, we aim to achieve more effective and visually appealing results in low-light scenarios. Obtaining pairs of low/normal-light videos, with motions, is more challenging than still images, which raises technical issues and poses the technical route of unpaired learning as a critical role. In addition, existing enhancement techniques often rely on pixel-level adjustments, which may fail to account for the semantic richness of the content. Consequently, these methods can produce results that lack the

clarity and detail necessary for effective analysis. We make endeavors in the direction of learning for low-light video enhancement without using defined Spatial-Temporal Compatible Learning [10] and unpaired ground truth [9]. Additionally, we emphasize the importance of gap-closing strategies in our work [11], particularly in the context of perceptual quality evaluation and optimization of low-light image enhancement. By addressing these challenges, we aim to improve the overall performance and visual quality of enhanced low-light videos.

**Future work** My prior research focused on addressing a key challenge in deep learning related to intelligent image and video enhancement, quality assessment, as well as image and video compression. In particular, I contributed to unpaired video enhancement, specifically targeting the major limitations of low-light video enhancement, which often relies heavily on the availability of ground truth data. My work involved developing novel algorithms that improve the quality of low-light video content without needing paired reference frames, thus enhancing usability in practical applications. Additionally, my research on generative image compression aims to reconstruct natural images under ultra-low bitrate constraints. This involves designing efficient generative models that can produce high-quality images while minimizing transmission cost, making them suitable for bandwidth-limited scenarios. By focusing on perceptual quality and compression efficiency, my work addresses the critical need for effective transmission and storage of visual data in various applications. Building on these and towards the goal of building an intelligent technique capable of understanding and interacting with the visual world, I aim to explore the following directions in the near future.

**Accelerating understanding and generation system.** We humans possess a holistic understanding of the visual world, allowing us to easily perceive the category, location, and shape of objects, and to interact with them. This fundamental capability is crucial for intelligent technologies that need to navigate and operate within visual environments. Reconstructing realistic scenes has significant applications in fields such as surveillance, robotics, virtual reality, and autonomous driving. While the scene understanding and generation have been extensively studied in China as a unified model, several key challenges remain. As deep learning models grow in complexity, the need for efficient inference solutions has become increasingly important, especially for practical applications. My goal is to develop a system that can fully understand and reconstruct scenes using world models with minimal human supervision. I plan to design an end-to-end model that simultaneously learns visual understanding alongside image and video reconstruction. To achieve this, I will investigate techniques for optimizing model architectures, such as pruning, quantization, and knowledge distillation, to reduce computational overhead without sacrificing accuracy. This aligns with my previous experience in image and video compression. I will also systematically

evaluate the state-of-the-art generative models for image and video generation for next-generation frameworks.

**Multimodal Foundation Models.** As we all know, humans really have this incredible way of understanding the world through all our senses—like seeing, hearing, touching, and even smelling. These senses work together and help us learn by comparing and aligning what we perceive. Imagine a future where intelligent agents, equipped with multiple sensors, could learn about the world just by interacting with it! This idea of multi-modal perception is super important, not just for how we think but also for real-world applications. To meet this challenge, I am committed to developing a multimodal foundation model that seamlessly integrates data from diverse sensors, learning to recognize the connections between them. Furthermore, I will collaborate with visionaries across disciplines to develop machine learning tools capable of unlocking the mysteries hidden within our multi-sensory world.

**Robustness Foundation Models.** While holistic scene understanding and generation have been extensively studied, several key challenges remain, particularly in the context of deep learning models that are growing increasingly complex. In addition, I will investigate metrics to evaluate the robustness of the model in diverse real-world scenarios, ensuring that these models maintain performance under varying conditions and disturbances. This work will not only enhance our theoretical understanding of robustness but also have practical implications for applications in critical domains such as autonomous systems.

**Multidisciplinary collaborations.** I have established collaborations with researchers in the industry, including Tencent Games and Alibaba DAMO Academy, to address both fundamental and applied research challenges related to generative models. In addition, I have collaborated with researchers from University of Trento, City University of Hong Kong, Lingnan University, Nanyang Technological University, University of Missouri, Communication University of China, Zhejiang University, Peking University, Jiangxi University of Finance and Economics, and South China Normal University to design AI algorithms. I am willing to commit to continuing these multidisciplinary collaborations and advancing the application of AI across various scientific domains.

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